

CS23532-COMPUTER NETWORKS

EXPERIMENT -1

Aim

Study the basic networking commands in Linux and Windows operating systems.

Objective

To learn and execute basic networking commands in Windows and Linux for viewing network configuration, testing connectivity, and troubleshooting network-related issues.

Theory

Networking commands help users obtain information about the network configuration, verify connectivity between devices, identify network problems, and monitor network communication. These commands are essential for network administrators and students working with computer networks.

Windows commands

```
C:\Users\dell>hostname dell
```

```
C:\Users\dell>ipconfig /all
```

Windows IP Configuration

Host Name dell
Primary Dns Suffix :
Node Type..... Mixed
IP Routing Enabled..... No
WINS Proxy Enabled No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description.....VirtualBox Host-Only Ethernet Adapter
Physical Address. : 0A-00-27-00-00-0E
DHCP Enabled..... No
Autoconfiguration Enabled Yes
Link-local IPv6 Address: fe80::aa4a:775:1de1:1757%14(Preferred)
IPv4 Address. : 192.168.56.1(Preferred)
Subnet Mask..... : 255.255.255.0

Default Gateway :

DHCPv6 IAID118095911

DHCPv6 Client DUID..... 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpi..... Enabled

Wireless LAN adapter Local Area Connection* 1:

Media StateMedia disconnected

Connection-specific DNS Suffix . :

Description.....Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. : F8-FE-5E-3A-70-56

DHCP Enabled..... Yes

Autoconfiguration Enabled Yes

Wireless LAN adapter Local Area Connection* 2:

Media StateMedia disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. : FA-FE-5E-3A-70-55

DHCP Enabled..... No

Autoconfiguration Enabled Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description.....Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. : F8-FE-5E-3A-70-55

DHCP Enabled Yes
Autoconfiguration Enabled Yes
Link-local IPv6 Address: fe80::b810:29c9:55e9:3a65%17(Preferred)
IPv4 Address. 192.168.1.100(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained : 26 August 2026 14:38:14
Lease Expires : 26 August 2026 15:38:11
Default Gateway : 192.168.1.1
DHCP Server.....: 192.168.1.1
DHCPv6 IAID167312990
DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
DNS Servers : 192.168.1.1
NetBIOS over Tcpi..... Enabled

Ethernet adapter Bluetooth Network Connection:

Media State Media disconnected
Connection-specific DNS Suffix . :
Description.....Bluetooth Device (Personal Area Network)
Physical Address. : F8-FE-5E-3A-70-59
DHCP Enabled..... Yes
Autoconfiguration Enabled Yes

Ethernet adapter Ethernet:

Media State Media disconnected
Connection-specific DNS Suffix . :
Description.....Realtek Gaming GbE Family Controller

Physical Address 2C-58-B9-DE-5F-5F

DHCP Enabled Yes

Autoconfiguration Enabled Yes

C:\Users\dell>arp -a

Interface: 192.168.56.1 ----0xe

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff	static

Interface: 192.168.1.100 --- 0x11

Internet Address	Physical Address	Type
192.168.1.1	4e-ec-85-ec-ff-15	dynamic
192.168.1.255	ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff	static

C:\Users\dell>netstat -r

=====

Interface List

14...0a 00 27 00 00 0e VirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56 Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55 Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59 Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5f Realtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

IPv4 Route Table

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.100	35
	192.168.1.0	255.255.255.0	On-link	192.168.1.100	291
	192.168.1.100	255.255.255.255	On-link	192.168.1.100	291
	192.168.1.255	255.255.255.255	On-link	192.168.1.100	291
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
	127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
	192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
	192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
	192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
	224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
	224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
	224.0.0.0	240.0.0.0	On-link	192.168.1.100	291
	255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
	255.255.255.255	255.255.255.255	On-link	192.168.56.1	281

255.255.255.255 255.255.255.255 On-link 192.168.1.100 291

Persistent Routes:

None

IPv6 Route Table

Active Routes:

If	Metric	Network	Destination	Gateway
----	--------	---------	-------------	---------

1	331	::1/128		On-link
---	-----	---------	--	---------

14	281	fe80::/64		On-link
----	-----	-----------	--	---------

17	291	fe80::/64		On-link
----	-----	-----------	--	---------

14	281	fe80::aa4a:775:1de1:1757/128		
----	-----	------------------------------	--	--

On-link

17	291	fe80::b810:29c9:55e9:3a65/128		
----	-----	-------------------------------	--	--

On-link

1	331	ff00::/8		On-link
---	-----	----------	--	---------

14	281	ff00::/8		On-link
----	-----	----------	--	---------

17	291	ff00::/8		On-link
----	-----	----------	--	---------

Persistent Routes:

None

C:\Users\dell>nslookup www.google.com Server:

UnKnown

Address: 192.168.1.1

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\dell>ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data:

Reply from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\dell>route print

=====

Interface List

14...0a 00 27 00 00 0e VirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56 Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55 Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59 Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5f Realtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

=====

IPv4 Route Table

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Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.100	35	
192.168.1.0	255.255.255.0	On-link	192.168.1.100	291	
192.168.1.100	255.255.255.255	On-link	192.168.1.100	291	
192.168.1.255	255.255.255.255	On-link	192.168.1.100	291	
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331	
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331	
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331	
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281	

192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	192.168.1.100	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	192.168.1.100	291

=====

Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network	Destination	Gateway
----	--------	---------	-------------	---------

1	331	::1/128		On-link
14	281	fe80::/64		On-link
17	291	fe80::/64		On-link
14	281	fe80::aa4a:775:1de1:1757/128		
				On-link
17	291	fe80::b810:29c9:55e9:3a65/128		
				On-link
1	331	ff00::/8		On-link
14	281	ff00::/8		On-link
17	291	ff00::/8		On-link

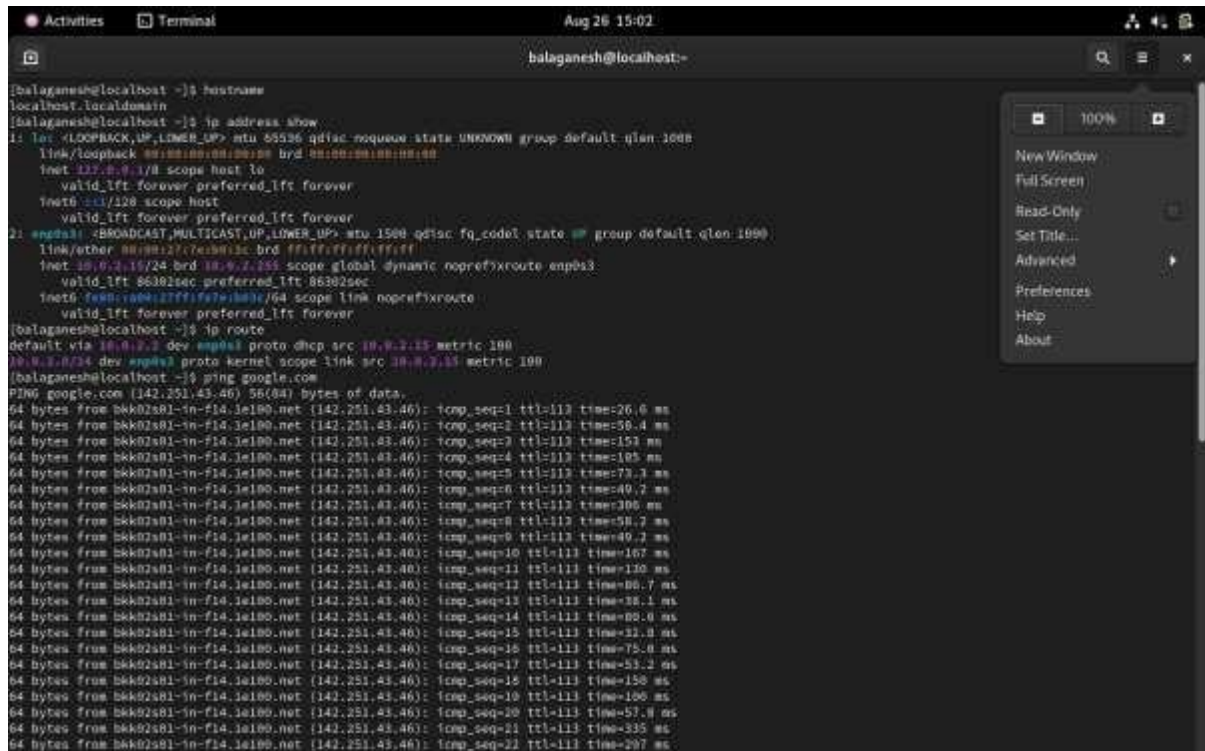
=====

Persistent Routes:

None

C:\Users\dell>tracert google.com

Linux commands



```
balaganesh@localhost:~$ hostname
localhost.localdomain
balaganesh@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:17:7c:b9:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute ens33
        valid_lft 86382sec preferred_lft 86382sec
    inet6 fe80::a00:27ff:fe7c:b9bc/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
balaganesh@localhost:~$ ip route
default via 10.0.2.2 dev ens33 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens33 proto kernel scope link src 10.0.2.15 metric 100
balaganesh@localhost:~$ ping google.com
PING google.com (142.251.43.46) 56(84) bytes of data:
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=1 ttl=113 time=26.6 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=2 ttl=113 time=50.4 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=3 ttl=113 time=151 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=4 ttl=113 time=105 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=5 ttl=113 time=77.3 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=6 ttl=113 time=49.2 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=7 ttl=113 time=306 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=8 ttl=113 time=58.2 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=9 ttl=113 time=49.2 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=10 ttl=113 time=167 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=11 ttl=113 time=130 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=12 ttl=113 time=80.7 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=13 ttl=113 time=38.1 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=14 ttl=113 time=80.0 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=15 ttl=113 time=32.0 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=16 ttl=113 time=75.0 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=17 ttl=113 time=53.2 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=18 ttl=113 time=158 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=19 ttl=113 time=186 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=20 ttl=113 time=57.8 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=21 ttl=113 time=335 ms
64 bytes from bkk02sbl-in-f14.1e100.net (142.251.43.46): icmp_seq=22 ttl=113 time=297 ms
```

Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route